

Reported earthquake depth uncertainty: geometry and source-estimator regimes

A compact robust geometry/reporting-regime equation predicts supplied depth uncertainty better than the original controls on held event clusters. The result concerns the catalog estimator, not independent true earthquake-depth accuracy or a new seismic law.

Data and evaluation

2607 July 2026 catalog events each have exactly one QuakeML origin. Positive, finite uncertainty/RMS/count eligibility yields 2274 development and 271 confirmation events, in 516/130 complete spatiotemporal clusters. Source coordinates/times define 100 km/7 day connected components before error analysis.

Target: **Author-reported origin depth uncertainty**, measured as km. Primary error: absolute natural-log ratio. Contemporaneous reported location-quality diagnostics, never pre-event earthquake forecast or independent true depth error.

Transitive clusters within 100 km great-circle chord and 7 days; complete cluster hashfolds. Distant events may still share network/systematic velocity model errors. Complete spatiotemporal connected clusters held out; same snapshot networks/measurement procedures. No new waveform or independent hypocenter calibration.

Findings and equations

$$\hat{\sigma} = \exp(a + b_R \ln R + b_N \ln N + c_G G + c_A A + c_D \ln(1 + |d|) + c_{10} I_{10} + c_S I_{<5})$$

The notation and all fitted coefficients are defined below.

P100-017-F01 - validated_extension (partially_supported). Reporting-regime and geometry covariates provide a compact predictor of author-supplied positive depth uncertainty in this source snapshot.

The selected robust log-uncertainty equation and its coefficients are given below.

Near-station leverage and angular coverage have known localization roles, but exact-depth indicators also encode catalog processing practices. Improvement cannot isolate equation structure from fitting loss without the disclosed later controls.

Falsifying evidence and limits: Worst cluster does not uniformly improve; cross-network uncertainty conventions, fixed-depth reporting and no independent truth limit mechanism claims. Matched robust controls were added only after first confirmation.

P100-017-F02 - informative_falsification (supported). Naive inverse-square-root phase-count scaling is insufficient as a universal uncertainty law here.

sigma proportional RMS/sqrt(Nphase) is the domain family; estimated free exponents/regimes in 002-006 challenge it.

Correlated arrivals, network geometry and differing source procedures violate independent identical measurements.

Falsifying evidence and limits: Reference predicts author formal uncertainty only; it cannot demonstrate true physical accuracy or hazard forecasting.

Selected executable model: attempt_004. $\sigma_{\text{hat}} = \exp(a + b_R \ln(\text{RMS}) + b_N \ln(\text{Nphase}) + c_G G + c_A A + c_D \ln(1 + |\text{depth}|) + c_{10} I(\text{depth}=10) + c_S I(\text{depth}<5))$, with $G = 0.5 \ln[1 + (d_{\text{min}}/(1 + |\text{depth}|))^2]$ and $A = -\ln(\max(1 - \text{gap}/360, 0.02))$. Numerical inputs use seconds, km and counts as declared fixed reference units.

Coefficients in that order=[0.5059683260395523, 0.38408763491978104, -0.1245459549330374, 0.08698001732673424, 0.3868077313593952, 0.36476450158184914, -0.6216509362540282, 0.08021397958958204]. This is phenomenological regression with dimensionless normalized numerical ratios; bN is not automatically the independent-phase exponent-0.5.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	0.3109677	0.2997433	1.755308
constant	0.7105668	0.7107532	2.576025
domain	0.6745367	0.6954668	1.566295
flexible	0.4301596	0.4341116	1.683547

Postfreeze diagnostic controls were fitted on development data after original confirmation exposure, without reselection. They address estimation-loss asymmetry; they are not fresh preregistered confirmation.

Added control	Development error	Exposed-cohort error
postfreeze_robust_domain	0.6632764	0.6813104
postfreeze_robust_flexible	0.4109096	0.4157607

Confirmation contains 271 scored observations in 130 groups. Individual-group scores and every attempted model remain available. Original confirmation log-ratioMAE 0.299743 versus constant 0.710753, information baseline 0.695467 and flexible 0.434112. Worst reference cluster 1.755308 exceeds domain 1.566295. After-exposure matched robust controls yield 0.681310 domain and 0.415761 flexible; these diagnostics keep reference unchanged and are not fresh confirmation.

Interpretation, limitations and use

One July 2026 catalog snapshot. Positive uncertainty reports only; mixed network estimation conventions, no universal standard-error confidence interpretation. Counts/geometry are outputs of location processing and causality is not identified. Whole-group errors/ranges; no row bootstrap or manufactured independent-population confidence. Confirmation becomes exposed after freeze.

Useful as a source-quality diagnostic and a benchmark for distinguishing physical inverse-information arguments from reporting/model conventions. No supplied confidence interval is validated against independently located events, waveforms or velocity structures.

Reproducibility and prior art

Run `python run.py` to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- Location quality and uncertainty definitions depend on contributing networks; not independent error truth. (checked 2026-10-02).

- Depth is often least-constrained, reference frames differ and fixed/negative depths occur. (checked 2026-10-02).
- Known importance of nearest-station distance relative to depth, phase counts and velocity model; inverse-information and geometry hypotheses are prior-inspired. (checked 2026-10-02).